

- 3 The β -globin gene codes for the β -globin polypeptide of haemoglobin. It has two alleles, **Hb^A** (normal) and **Hb^S** (sickle cell). The sickle cell allele differs from the normal allele due to a base substitution mutation and this mutation results in a single amino acid change to the β -globin polypeptide.

There are three possible genotypes and phenotypes.

- **Hb^S Hb^S**, sickle cell anaemia, a severe disease
- **Hb^A Hb^S**, sickle cell trait with mild or no symptoms of sickle cell anaemia
- **Hb^A Hb^A**, normal (healthy)

A man and woman who both have sickle cell trait may choose to have children by IVF. This allows the genotype of embryos to be determined by gene testing before the embryos are implanted. Embryos with the normal genotype can then be selected and implanted into the mother.

One technique that can be used in gene testing an embryo for the **Hb^S** allele is restriction fragment length polymorphism (RFLP) analysis. This involves digesting a DNA sample from an embryo with a restriction endonuclease and then separating the DNA fragments by gel electrophoresis. The position of the DNA fragments on the gel can show if the embryo has the **Hb^S** allele.

- (a) The first step in testing an embryo for the **Hb^S** allele by RFLP analysis requires many copies of the part of the β -globin gene in which the mutation causing sickle cell anaemia occurs.

- (i) Name the technique used to produce many copies of a DNA sequence from a very small quantity of DNA.

.....[1]

- (ii) Explain why it is necessary to copy this DNA sequence many times in order to test embryos for **Hb^S** alleles by RFLP analysis.

.....
.....[1]

The restriction site for *Mst*II is shown in Fig. 3.1.



Hb^A



Hb^S



- [illegible]

- (c) After cutting with *Mst*II, the DNA fragments are separated by gel electrophoresis.

Explain how gel electrophoresis separates DNA fragments cut with restriction endonucleases.

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.....[3]

- (d) Four embryos, **1**, **2**, **3** and **4**, were tested for the **Hb^S** allele using RFLP analysis. Fig. 3.4 shows the DNA fragments separated by gel electrophoresis for the four embryos. The DNA fragments for two individuals of known genotype, homozygous for **Hb^A** and homozygous for **Hb^S**, are also shown.

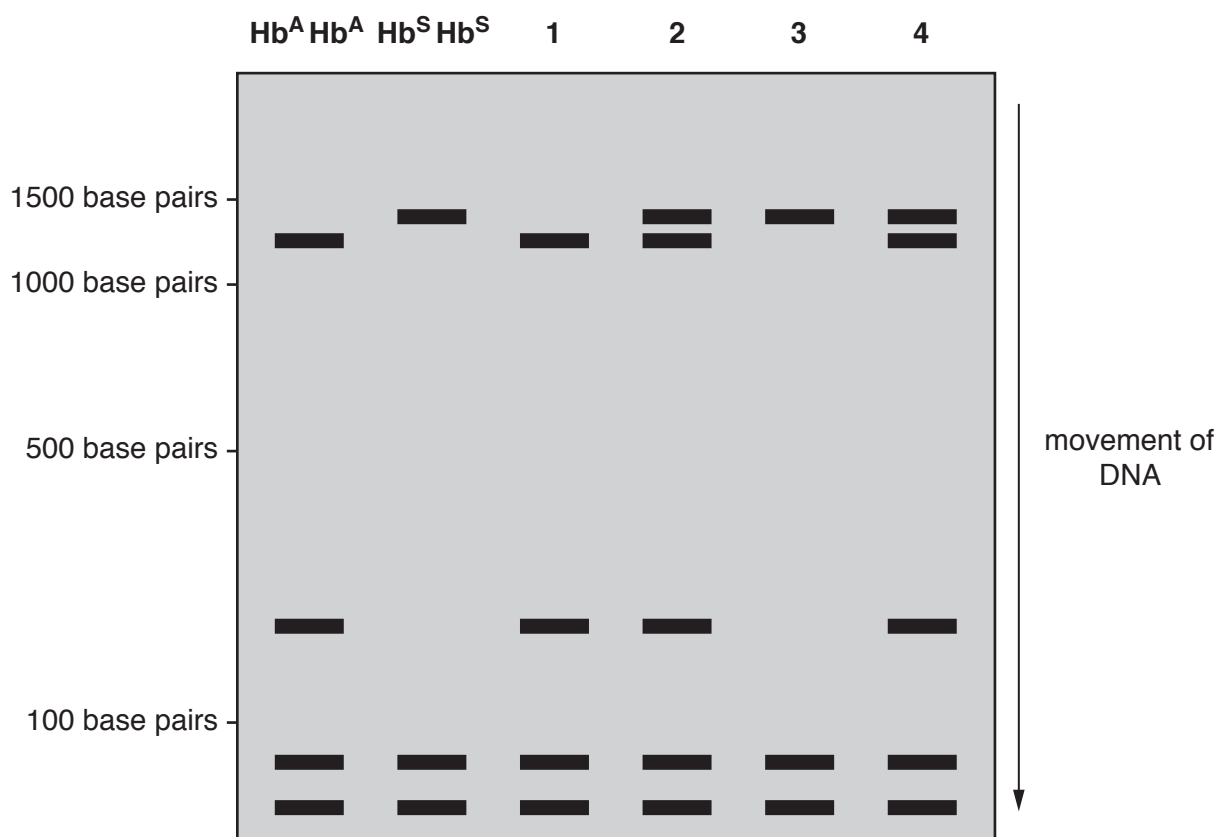


Fig. 3.4

(i) State the purpose of using DNA from individuals homozygous for **Hb^A** and for **Hb^S**.

.....
[1]

(ii) With reference to Fig. 3.4, complete Table 3.1 to show the genotypes of embryos **2**, **3** and **4**.

Table 3.1

embryo	genotype
1	Hb^A Hb^A
2	
3	
4	

[2]

(e) Discuss the ethical and social considerations of gene testing embryos for genetic diseases.

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[3]

[Total: 15]